

Unsupervised Learning - Lab 06 - Clustering Comparison

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Agenda

- Motivation:
 - This sixth lab session aims to compare and validate two different solutions of clustering (k-means, hierarchical) applied to an unknown matrix of data (with low dimensionality and continuous attributes).
- Steps:
 - In this lab, you are going to (re-)use already developed code.
 - Check solutions of Lab 03, Lab 04, Lab 05.
 - Moreover, some new strategies for validating clustering solutions will be used.
 - Both hierarchical and k-means will be applied, with their own validation.
 - Finally, comparison will be operated to guess the clusters, and cross-check with the true labels (given at the end of the session).

For k-means, two different strategies will be implemented:

1. Run hierarchical clustering, find a candidate number of clusters, then run k-means with that given number.
2. Use the elbow method and find the best number of clusters only using k-means.

- The data to work on will be available on eLearning at the beginning.
- The true labels to cross-check your clustering solutions will be available on eLearning in a separate.